	SAMRAT ASHOK TECHNOLOGICAL INSTITUTE (DEGREE) VIDISHA (M.P)
Department	Cybersecurity &IoT
Branch	CSE- Cybersecurity
Year & Sem	B.Tech. First Year (II - Semester)
Session	Batch 2026-27 onwards

Category of Course	**Course Title**	**Course Code**				
			L	**T**	**P**	**Credits – 1**
ETC	Computer Workshop (LINUX)	CY26103P	0	0	2	

Course Code	Course Title	**Maximum Marks Allotted**					**Total Marks**
		Theory Paper			Practical		
		End Sem.	Mid Sem. MST	Quiz/ Assignment	End Sem.	Lab Work	
CY26103P	Computer Workshop (LINUX)	--	--	--	30	20	50

Course Objectives	• To teach principles of operating system including File handling utilities, security by file permissions, Process utilities Command, Basic Linux commands, Scripts and filters. • To familiarize fundamentals of the Bourne again shell (bash), shell programming, pipes, input and output red arithmetic in shell interrupt processing, functions, debugging shell scripts. • To impart fundamentals of file concepts kernel support for file, Files structure related system calls (file API'S) • To facilitate students in understanding Inter process communication. • To facilitate students in understanding semaphore and shared memory. • To facilitate students in understanding process.
Prerequisite Knowledge	Fundamentals of computers
Course Description	The Linux Computer Workshop introduces students to the fundamentals of the Linux operating system and command-line environment. Participants will learn basic Linux commands, file and directory management, user permissions, shell scripting, and system utilities through hands-on practice. The workshop emphasizes practical skills for system navigation, software management, and basic administration tasks.
Course Outcomes	**Upon completion of the course, student will be able to** 1. Use various Linux commands that are used to manipulate system operations at admin level and a prerequisite to pursue job as a Network administrator. 2. Write Shell Programming using Linux commands. 3. Design and write application to manipulate internal kernel level Linux File System. 4. Develop IPC-API's that can be used to control various processes for synchronization. 5. Develop Network Programming that allows applications to make efficient use of resources available on different machines in a network.

List of Experiments

1.	Demonstrate a program using echo, printf, script, passwd, uname, who, date, stty, pwd commands.	CO1
2.	Illustrate a program using unlink, du, df, mount, umount, find, unmask, ulimit, ps, w, finger, arp, ftp commands.	CO1
3	Evaluate a program using telnet, rlogin.Text Processing utilities and backup utilities, tail, head, sort, nl, uniq, grep, egrep, fgrep, cut, paste, join, tee, pg, comm, cmp, diff, tr, awk commands.	CO2
4	Interpret a shell script that accepts a file name, starting and ending line numbers as arguments and displays all the lines between the given line numbers.	CO2
5	Illustrate by writing script that will print, message —Hello World, in Bold and Blink effect, and in different colors like red, brown etc using echo commands?	CO2
6	Develop shell script that deletes all lines containing a specified word in one or more files supplied as arguments to it.	CO3
7	Illustrate by writing script using for loop to print the following patterns?	CO3
8	Illustrate a shell script that displays a list of all the files in the current directory to which the user has read, write and execute permissions.	CO4
9	Demonstrate a program inter-process communication.	CO5
10	Create a program to communicate using sockets.	CO5